

Ayush Urs

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EDUCATION

University of California, San Diego

San Diego, CA

M.S. Computer Science

University of California, San Diego

San Diego, CA

B.S. Mathematics – Computer Science

GPA: 3.89 September 2023 – March 2026

Relevant Coursework: Machine Learning: Algorithms, Data Structures, Stochastic Processes, Recommender Systems,

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, Dart, MATLAB, Stata

Frameworks: TensorFlow, PyTorch, Keras, LangChain, scikit-learn, pandas, NumPy, Apache Spark, Databricks, Flutter

Tools: Cursor, MCP(Model Context Protocol), CI/CD, CUDA, AWS (EC2, ECS, CloudWatch), GCP, Vertex AI, PostgreSQL, Nautilus, MongoDB, LoRA, DPO, SQLite, Docker, Git, VizTracer, HPC, Jupyter, VS Code, Claude Code

EXPERIENCE

AI Researcher

Remote

University of Wisconsin–Madison

June 2026 – Present

- Architected a **7-agent LangChain pipeline** decoupling literature parsing and execution for automated simulation generation, cutting manual debugging time by **38%** and accelerating experimentation cycles by **30%**.
- Standardized tool access via a scoped **MCP server**, added **AST**-based validation and **CI/CD** regression testing, cutting integration overhead **41%**, failed runs **35%**, and catching **63+**% of breaking changes before execution.

Machine Learning Researcher

San Diego, CA

Halcioğlu Data Science Institute

March 2026 – June 2026

- Designed a novel **bandit-based selection** strategy with **divide-and-conquer decomposition** for **LLM-driven** algorithm discovery. Allocated compute more efficiently toward promising solution candidates than existing frameworks (**SkyDiscover**, **EoH**, **FunSearch**).
- Evaluated the strategy across **200+** optimization tasks, achieving **31%** fewer **LLM** calls and **24%** faster convergence, orchestrating program generation over a **SQLite** strategy database with **unit tests** and **VizTracer**.

Machine Learning Researcher

San Diego, CA

Human-centered eXtended Intelligence (UCSD Lab)

September 2025 – Present

- Built Reddit-based pipelines to train LLMs toward regulated responses across **CERQ**(cognitive emotion regulation) using **Vertex AI** for preference-pair generation, improving classification by **21** percentage points.
- Fine-tuned **transformer models** on **100K+** samples using **DPO** on **4x A100** Nautilus clusters, with responses preferred in **72%** of human evaluations for emotional regulation across regulated and dysregulated variants.

Machine Learning Intern

New York, NY (Hybrid)

Myna Voice Labs (Research Collaboration with Princeton PhD Scholars)

August 2025 – April 2026

- Achieved **89%** accuracy with strong cross-linguistic generalization by developing Parkinson's disease detection models from speech using **SVMs**, **CNNs**, and **Audio Spectrogram Transformers (AST)**.
- Confirmed model robustness with a **20–28x** separation in confidence interval width between class predictions by evaluating audio perturbations and noise using **Bayesian intervals** across **OpenL3** and **Whisper** feature representations; integrated **LoRA-based continual learning** to retain **95.6%** of prior-task performance.

PROJECTS

Skin Lesion Detection System | *PyTorch, YOLOv8, Docker, Kubernetes*

- Contributed to an end-to-end skin lesion classification pipeline on **HAM10000**, benchmarking a custom CNN against **EfficientNet-B0**, **ResNet18**, and **YOLOv8s**, reaching **81.75%** top-1 accuracy.
- Built **GPU training infrastructure** with **Docker** and **Kubernetes** on **A100** nodes, supporting multi-GPU training, persistent checkpoint storage, and interactive debugging pods.

RAG Pipeline for Internal Query Answering | *LangChain, sentence-transformers, MongoDB, FastAPI, AWS*

- Built **RAG** pipeline to replace manual internal doc search, using **LangChain** and **sentence-transformers**.
- Designed data layer with **MongoDB** for embeddings and document storage, orchestrated via **FastAPI**, containerized with **Docker**, and deployed on **AWS**, sustaining **100+** requests at **p95 latency of 1-2 seconds**.

AI-Integrated Island for Eutrophication Control | *Reinforcement Learning, Python* | [Research Paper](#)

- Developed AI island research prototype using Bellman equation targeting high-phosphate regions; **75%** reduction.